Literature answer: Gross' Brownian motion fails the polarity principle

Status: literature_already_answered (negative answer)

Original open problem

L. Beznea, A. Cornea, and M. Röckner, *Potential theory of infinite dimensional Lévy processes*, arXiv:1007.2379v2, Section 4.8, page 19, immediately after Remark 4.15, ask:

> Does the axiom of polarity hold for infinite-dimensional Brownian motion?

Here the axiom is Hunt's Hypothesis (H): every semipolar Borel set is polar.

Decisive supporting paper

P. J. Fitzsimmons, *Gross' Brownian Motion Fails to Satisfy the Polarity Principle*, Rev. Roumaine Math. Pures Appl. **59** (2014), no. 1, 87–91. Theorem 1, on journal page 87 (PDF page 1), states that Gross' Brownian motion does not satisfy the axiom of polarity. Proposition 1, on journal page 88, supplies a Borel set that is semipolar but not polar.

Identification

The supporting paper explicitly identifies the same question. Its opening paragraph cites the Beznea–Cornea–Röckner paper, says that it left open whether Gross' process satisfies the axiom of polarity, and announces a counterexample.

The mechanism is an intrinsic clock. On the classical Wiener state space $E = C([0,1];\mathbb{R})$, Fitzsimmons defines the quadratic-variation functional

$$Q(x) = \lim_{n\to\infty} \sum_{k=1}^{2^n} \big(x(k2^{-n}) - x((k-1)2^{-n})\big)^2$$

when the limit exists. The shell $S = \{Q = 1\}$ is not polar because a process started at the origin belongs to S at time one almost surely. Proposition 2 proves

$$Q(X_t) = Q(X_0) + t \qquad (t > 0),$$

whenever $Q(X_0) < \infty$. Consequently the process visits S at most once; the strong Markov argument in the paper makes S semipolar. Thus S is the required semipolar, nonpolar counterexample.

Scope and provenance

The formal theorem is stated for Gross' Brownian motion on the classical Wiener space. This already disproves general validity of the polarity axiom for infinite-dimensional Brownian motion. Fitzsimmons also states that the construction can be modified for a general abstract Wiener space, but does not spell out that extension. The paper does not claim the analogous result for the broader class of infinite-dimensional Lévy processes.

This is an explicit later-paper answer, not a new result of the present run. The identification was confirmed from the original page-19 question, the supporting paper's Theorem 1 and Propositions 1–2, an exact-phrase search for "axiom of polarity" with "infinite dimensional Brownian motion", and the 2019 survey on Hunt's Hypothesis (H).

References

[1] L. Beznea, A. Cornea, and M. Röckner, *Potential theory of infinite dimensional Lévy processes*, J. Funct. Anal. **261** (2011), 2845–2876; arXiv:1007.2379.

[2] P. J. Fitzsimmons, *Gross' Brownian Motion Fails to Satisfy the Polarity Principle*, Rev. Roumaine Math. Pures Appl. **59** (2014), no. 1, 87–91.

[3] Z.-C. Hu and W. Sun, *Hunt's Hypothesis (H) for Markov Processes: Survey and Beyond*, arXiv:1908.06825.